

Numerical Analysis of Personalized PageRank Using Variance-Reduced Monte Carlo Methods on Football Passing Networks

Based on Liao et al., SIGMOD 2023

Member 1 ID: 2105106

Member 2 ID: 2105108

Member 3 ID: 2105109

Member 4 ID: 2105112

Member 5 ID: 2105120

Course: CSE 402 - Numerical Analysis **Sec:** B / B2

“Efficient Personalized PageRank Computation: The Power of Variance-Reduced Monte Carlo Approaches”

Liao et al. • *SIGMOD 2023*

Base Paper Overview - Continue

Problem

Monte Carlo estimation of Personalized PageRank (PPR) can require many random-walk samples and may suffer from high estimator variance.

Core Idea

Investigate efficient PPR computation via Monte Carlo random walks combined with a **variance-reduction** strategy.

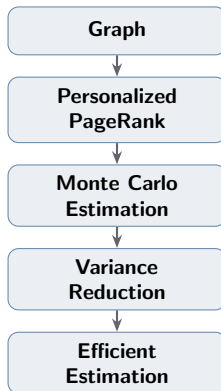
Numerical Connection

PPR can be computed by **power iteration** (deterministic) or **Monte Carlo random walks** (stochastic) — both central topics of Numerical Analysis.

Why it Fits CSE 402

Iterative computation • convergence analysis • approximation error • stochastic estimation • variance • computational efficiency

Base Paper Overview - Continue



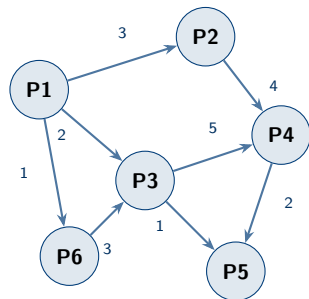
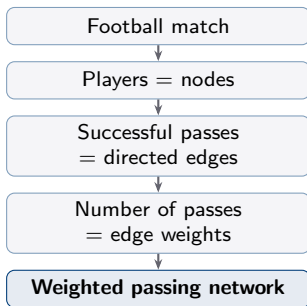
Our Direction

Reproduce the core methodology,
then apply it to football passing networks.

Problem Statement

Research Question

How can Personalized PageRank be computed **accurately and efficiently** on football passing networks, and how does **variance-reduced Monte Carlo** compare with standard Monte Carlo and deterministic Power Iteration?



Player-passing network

Goal: measure structural influence within the passing network

Problem Scope

Project Scope

Stage 1 — Base Paper Reproduction

- Implement PPR using the **Power Method**
- Implement **Standard Monte Carlo** estimation
- Implement **Variance-Reduced Monte Carlo** (per base paper)
- Validate on standard graph benchmarks

Stage 2 — Football Application

- Extract pass events from **StatsBomb Open Data**
- Construct player-to-player weighted directed graphs
- Compute Personalized PageRank on match networks

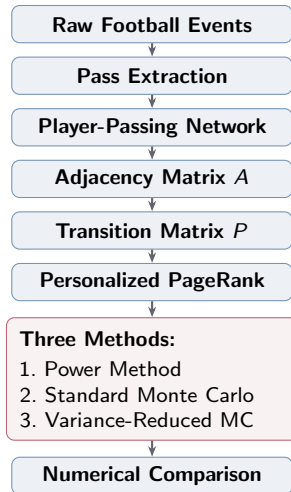
Stage 3 — Numerical Analysis

- Compare accuracy, convergence, variance, runtime
- Analyse sample efficiency and ranking stability

Out of Scope

Not intended to predict match outcomes or determine the absolute “best” player.

Expected Methodology



Key Mathematical Formulation

PPR Power Iteration:

$$\mathbf{r}^{(k+1)} = \alpha P \mathbf{r}^{(k)} + (1 - \alpha) \mathbf{s}$$

Convergence criterion:

$$\|\mathbf{r}^{(k+1)} - \mathbf{r}^{(k)}\| < \varepsilon$$

α : damping factor; $\mathbf{s}$: seed/personalization vector; ε : tolerance

Evaluation Metrics

- Accuracy (L_1/L_2 error)
- Convergence rate
- Monte Carlo variance
- Runtime
- Samples required
- Ranking stability

THANK YOU